Lecture 1: Markov processes and Diffusions

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Outline

Regular Conditional Distribution

Markov Processes and Diffusions

RCD

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. $X : (\Omega, \mathcal{F}) \to (E, \mathcal{E})$ a measurable map, and $\mathcal{G}$ a σ-field $\subseteq \mathcal{F}$.

- When $E = \mathbb{R}$, we define the **conditional expectation** of X given $\mathcal{G}$, $\mathbf{E}(X|\mathcal{G})$, to be any random variable Y that satisfies
  (a) $Y \in \mathcal{G}$;
  (b) for all $A \in \mathcal{G}$, $\mathbf{E}(X; A) = \mathbf{E}(Y; A)$.
- $Q_{\mathcal{G}} : \Omega \times \mathcal{E} \to [0, 1]$ is said to be a **regular conditional distribution** for X given $\mathcal{G}$ if
  (a) For each $A \in \mathcal{E}$, $\omega \mapsto Q_{\mathcal{G}}(\omega, A)$ is a version of $\mathbf{E}(\mathbf{1}_A(X)|\mathcal{G})$;
  (b) For a.e. $\omega \in \Omega$, $A \mapsto Q_{\mathcal{G}}(\omega, A)$ is a probability measure.

If $E = \Omega$, $X(\omega) = \omega$, then $Q_{\mathcal{G}}$ is called a **regular conditional probability**.

RCD

Proposition 1

Let $Q_{\mathcal{G}}$ be a RCD for X given $\mathcal{G}$. If $f : E \to \mathbb{R}$ has $\mathbf{E}|f(X)| < \infty$, then

$$\mathbf{E}(f(X)|\mathcal{G})(\omega) = \int_E f(x) Q_{\mathcal{G}}(\omega, \mathrm{d}x) \quad a.s..$$

Theorem 1 (cf. [Dur19])

RCD exists if E is a Polish space and $\mathcal{E} = \mathcal{B}(E)$.

Transition Probability

A **transition kernel** K on a measurable space $(E, \mathcal{E})$ is a map $K : E \times \mathcal{E} \to \mathbb{R}_+$ such that for each $x \in E$, $K(x, \cdot)$ is a measure, and for each $A \in \mathcal{E}$, $K(\cdot, A)$ is measurable. If, furthermore, $K(x, E) = 1$ for all $x \in E$, then K is a **transition probability**.

Theorem 2 (cf. [Dur19])

Suppose X and Y take values in a Polish space E. There is a transition probability $P : E \times \mathcal{E} \to [0, 1]$ such that $(\omega, A) \mapsto P(Y(\omega), A)$ is a RCD for X given Y, and P is unique a.s. $\mathrm{law}(Y)$.

We denote $P(y, A)$ by $\mathbf{P}(X \in A | Y = y)$.

Markov Processes

Let $\mathbb{F} = (\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geqslant 0}, \mathbf{P})$ be a filtered probability space. Let $\underline{X} = \{X_t\}_{t \geqslant 0}$ be a process takes values in $(E, \mathcal{E})$. The **Markov property** of $\underline{X}$ then says that, for any times $s \leqslant t$ and $A \in \mathcal{E}$,

$$\mathbf{P}(X_t \in A | \mathcal{F}_s) = \mathbf{P}(X_t \in A | X_s), \quad \mathbf{P} - a.s.. \tag{1}$$

Exercise 1

(1) *is equivalent to*

$$\mathbf{E}(F(X_{s+\cdot}) | \mathcal{F}_s) = \mathbf{E}(F(X_\cdot) | X_s) \quad \mathbf{P} - a.s., \tag{2}$$

where $F : E^{\mathbb{R}_+} \to \mathbb{R}$ *is a bounded measurable function.*

CK-equation

In this course, we always assume E is a Polish space. By Theorem 2, one can define

$$P_{s,t}(x, A) := \mathbf{P}(X_t \in A | X_s = x), \quad s \leqslant t.$$

For each bounded measurable function $f : E \to \mathbb{R}$, set

$$P_{s,t} f(x) := \int_E P_{s,t}(x, \mathrm{d}y)\, f(y) = \mathbf{E}(f(X_t)|X_s = x).$$

Noting that

$$\begin{aligned} P_{r,t} f(X_r) =& \mathbf{E}(f(X_t)|X_r) = \mathbf{E}\left[\mathbf{E}\left(f(X_t)|X_s, X_r\right)|X_r\right] \\ =& \mathbf{E}\left[\mathbf{E}\left(f(X_t)|X_s\right)|X_r\right] \\ =& \mathbf{E}(P_{s,t} f(X_s)|X_r) = P_{r,s} P_{s,t} f(X_r), \end{aligned}$$

we have

$$P_{r,t}(\cdot, \mathrm{d}y) = \int_E P_{r,s}(\cdot, \mathrm{d}z) P_{s,t}(z, \mathrm{d}y) \quad \mathrm{law}(X_r) - a.s.$$

Time-homogeneous

Therefore, a Markov process can be described by a collection of transition probabilities $P_{s,t}$. In this course, we only consider the time-homogeneous case, meaning that $P_{s,t}$ depends only on the size $t-s$.

A homogeneous transition function on $(E, \mathcal{E})$ is a collection $\{P_t\}_{t \geqslant 0}$ of transition probabilities on $(E, \mathcal{E})$ such that $P_s P_t = P_{s+t}$ for all $s, t \geqslant 0$. A process $\underline{X}$ is Markov with transition function P_t, and with respect to a filtered probability space $\mathbb{F}$ if it is adapted and $\mathbf{E}(f(X_{t+s})|\mathcal{F}_s) = P_t f(X_s)$. One can verify that

$$\mu_{t_1,\cdots,t_k}(A_1 \times \cdots \times A_k) := \mathbf{P}(X_{t_1} \in A_1, \cdots, X_{t_k} \in A_k)$$
$$= \int_{\mathbb{R}^d \times A_1 \times \cdots \times A_k} \mu_0(\mathrm{d}z_0) P_{t_1}(z_0, \mathrm{d}z_1) \cdots \mathrm{d}z_{k-1}) P_{t_k - t_{k-1}}(z_{k-1}, \mathrm{d}z_k).$$

Construction of Markov processes

Question: Given a collection of transition probabilities $\{P_t\}_{t\geqslant 0}$, how to construct the corresponding Markov process?

The following Kolmogorov consistency theorem gives an affirmative answer to this.

Theorem 3 (Kolmogorov consistency theorem)

Let E be a Polish space. For each $k \in \mathbb{N}$ and $t_1, \cdots, t_k \in I$, $\mu_{t_1,\cdots,t_k}$ is a probability measure on E^k. Assume that $\mu_{\cdots}$s satisfy two consistency conditions

(a) for all measurable sets $A_i \subset E$,

$$\mu_{t_1\ldots t_k}(A_1 \times \cdots \times A_k) = \mu_{t_1\ldots t_{k+m}}\Big(A_1 \times \cdots \times A_k \times \underbrace{E \times \cdots \times E}_{m}\Big);$$

Construction of Markov processes

(b) *for each permutation* π *of* $\{1, \cdots, k\}$ *and for all measurable sets* $A_i \subset E$,

$$\mu_{t_{\pi(1)} \ldots t_{\pi(k)}} \left(A_{\pi(1)} \times \cdots \times A_{\pi(k)} \right) = \mu_{t_1 \ldots t_k} \left(A_1 \times \cdots \times A_k \right).$$

Then there exists a probability measure μ *on* $\Omega = E^I$ *such that*

$$\mu(\omega_{t_1} \in A_1, \cdots \omega_{t_k} \in A_k) = \mu_{t_1 \cdots t_k}(A_1 \times \cdots \times A_k).$$

i.e. under μ *the canonical process* $\underline{\omega}$ *has* $\mu_{t_1 \ldots t_k}$ *as its finite-dimensional distributions relative to times* $t_1 \ldots t_k$.

Example: Brownian Motion

As an example, standard Brownian motion, W, has the defining property that $W_t - W_s$ is normal with mean 0 and variance $t - s$ independently of $\{W_u \colon u \leqslant s\}$, for times $s < t$. Equivalently, it is a Markov process with the transition function

$$P_t f(x) = \frac{1}{(2\pi t)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \mathrm{e}^{-\frac{|y-x|^2}{2t}} f(y)\,\mathrm{d}y.$$

Infinitesimal generators

Let $E = \mathbb{R}^d$. Assume that the following limit

$$\mathcal{L}\varphi(x) = \lim_{t \to 0} \frac{1}{t} \int_{\mathbb{R}^d} (\varphi(y) - \varphi(x)) P_t(x, \mathrm{d}y) \tag{3}$$

exists for all $\varphi \in C_c^\infty$, and that $P_t(x, \mathrm{d}y) = p(t, x, y)\mathrm{d}y$. Under mild conditions, this implies

$$\frac{\mathrm{d}}{\mathrm{d}t} P_t f = \mathcal{L} P_t f = P_t \mathcal{L} f, \quad \forall f \in C_c^\infty$$

and

$$\begin{cases} \partial_t p(t, x, y) = \mathcal{L} p(t, \cdot, y)(x) \\ p(0, \cdot, y) = \delta_y \end{cases} \tag{4}$$

in the sense of distribution.

Infinitesimal generators

Denote the space of all real-valued functions on $\mathbb{R}^d$ by $M(\mathbb{R}^d)$. Assume that $\varphi(x_0) = \sup_{x\in\mathbb{R}^d} \varphi(x)$, then

$$\mathcal{L}\varphi(x_0) = \lim_{t\to 0} \frac{1}{t} \int_{\mathbb{R}^d} (\varphi(y) - \varphi(x_0)) P_t(x_0, \mathrm{d}y) \leqslant 0.$$

We say that an operator $L : D(L) \subseteq M(\mathbb{R}^d) \to M(\mathbb{R}^d)$ satisfies the **positive maximum principle** on $D(L)$ if for all $\varphi \in D(L)$ such that

$$L\varphi(x_0) \leqslant 0,$$

if $\varphi(x_0) = \sup_{x\in\mathbb{R}^d} \varphi(x)$.

Infinitesimal generators

Theorem 4 (Courrège theorem, see [Cou68])

Let $L : C_c^\infty(\mathbb{R}^d) \to M(\mathbb{R}^d)$ be a linear operator. Then L satisfies the positive maximum principle iff there exist, uniquely determined by L for each $x \in \mathbb{R}^d$,

(i) *a nonnegative definite symmetric matrix $(a_{ij})_{1\leqslant i,j\leqslant d}$;*

(ii) *a vector $b(x) \in \mathbb{R}^d$;*

(iii) *a $c(x) \geqslant 0$;*

(iv) *a Borel measure $\nu(x, \mathrm{d}z)$ on $\mathbb{R}^d\backslash\{0\}$ satisfying $\int_{\mathbb{R}^d\backslash\{0\}}(1 \wedge |z|^2)\nu(x, \mathrm{d}z) < \infty$*

such that

$$\begin{aligned} L\varphi(x) &= \frac{1}{2}a_{ij}(x)\partial_{ij}\varphi(x) + b_i(x)\partial_i\varphi(x) - c(x)\varphi(x) \\ &+ \int_{\mathbb{R}^d\backslash\{0\}}(\varphi(x+z) - \varphi(x) - \mathbf{1}_{B_1}(z)z \cdot \nabla\varphi(x))\nu(x, \mathrm{d}z). \end{aligned} \tag{5}$$

Infinitesimal generators

Remark 1

1. *Suppose that $\mathcal{L}$ is defined by* (3)*, then it satisfies the positive maximum principle and $\mathcal{L}$ has representation* (5) *with $c = 0$;*
2. *L in* (5) *is the generator of a* **Lévy process** *if $c = 0$ and a, b and ν are independent with x.*

In the rest of this chapter, we add the assumption that the operator $\mathcal{L}$ in (3) is **local**, in the sense that $\mathcal{L}\varphi(x) = 0$ whenever φ vanishes in a neighborhood of x. Roughly speaking, local infinitesimal generators correspond to Markov processes with continuous trajectories (**Diffusion processes**).

Exercise 2

The operator $\mathcal{L}$ defined in (3) is local iff

$$\mathcal{L}\varphi(x) = \frac{1}{2}a_{ij}(x)\partial_{ij}\varphi(x) + b_i(x)\partial_i\varphi(x), \tag{6}$$

or

$$\lim_{t\to 0}\frac{1}{t}P_t(x, B_r^c(x)) = 0.$$

Exercise 3

Let X be a Markov process on $\mathbb{R}^d$ with transition function P_t. If

$$\lim_{u\downarrow 0}\frac{1}{u}\sup_{t\in[0,u];x\in\mathbb{R}^d} P_t(x, B_r^c(x)) = 0,$$

then there is a continuous Markov process Y defined on same probability space such that $\mathbf{P}(X_t = Y_t) = 1$ for each $t \geqslant 0$.

Kolmogorov's construction of diffusion processes

Question: given a and b, how to construct a diffusion process satisfies (6)*?*
Kolmogorov's idea was to recover P_t from (4).

Theorem 5 (see [Fri08])

Assume that $b \in L^\infty$*, and* $a \in C^\alpha$ *and there is a constant* $\lambda \in (0,1)$ *such that* $\lambda|\xi|^2 \leqslant a^{ij}\xi_i\xi_j \leqslant \lambda^{-1}|\xi|^2$*. There is a unique continuous fucntion* $p : (0,T) \times (\mathbb{R}^d \times \mathbb{R}^d \backslash \Delta) \to \mathbb{R}_+$ *such that for each* $t > 0$ *and each* $x, y \in \mathbb{R}^d$*.*

$$\partial_t p_t(x,y) = \mathcal{L} p_t(\cdot, y)(x), \tag{7}$$

Kolmogorov's construction of diffusion processes

Moreover, p fulfils

(a) *(Two-sided estimate)*

$$t^{-d/2}\mathrm{e}^{-|x-y|^2/Ct} \lesssim p_t(x,y) \lesssim t^{-d/2}\mathrm{e}^{-C|x-y|^2/t};$$

(b) *(Conservative)*

$$\int_{\mathbb{R}^d} p_t(x,y)\mathrm{d}y = 1;$$

(c) *(C-K equation)*

$$p_{t+s}(x,y) = \int_{\mathbb{R}^d} p_t(x,z)p_s(z,y)\mathrm{d}z.$$

Question

How to construct diffusion processes using probabilistic methods?

References

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